

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Wenlin Zhang¹, Sam Lin^{1,6}, Shuai Li³, Ye Yuan¹,
Zheng Wang^{1,2}, Ziteng Ni⁴, Heng Kang Yao^{2,4}, Yongcheng Li⁵, Davide Momi⁷

Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Given the available ECG/heartbeat reference, we propose a repetitive-control framework that models cardiac artifact as a heartbeat-aligned periodic disturbance. Each EEG sample is assigned a cardiac phase index wrapped around ECG R-peaks; phase-synchronous averaging then preserves the heartbeat-correlated component while attenuating non-phase-locked activity. Subtracting this phase-indexed average yields effective artifact removal in the non-causal offline setting. For online real-time operation, an iterative-learning estimator is used to track the disturbance estimate in either a same-beat predictive mode (strict causal) or a one-beat-delayed mode (buffered causal). The non-causal and buffered modes are evaluated on behind-the-ear EEG from a public wearable epilepsy dataset and on semi-synthetic data in which reference predictability is swept from linear to strongly nonlinear.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. As one of the primary contaminations in wearable ear-EEG, CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2].

Classical ECG-referenced adaptive cancellation is well established [3]–[5], and it is a natural baseline. In our target regime it faces three practical limits: earables are channel-starved for blind source separation [6], learned denoisers are benchmarked mainly on high-density montages [7], and—most restrictive for a wearable deployment—reference-driven cancellation must be driven by the ECG *waveform*, sample-aligned to the EEG. Our real-data observations sharpen this last limit rather than dismissing it: after beat-synchronous pooling, the ear-EEG artifact retains appreciable coherence with the ECG over the lower part of the band, so an

aligned waveform reference does carry usable information. The obstacle is obtaining one. A companion wearable exposes a heartbeat event stream rather than a synchronized ECG trace, wrist PPG carries no ECG waveform at all, and cross-device latency and jitter are unknown and time-varying. Beat timing, by contrast, is a single scalar per cardiac cycle that crosses the device boundary intact, and the artifact remains phase-locked to it. This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing. Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [8], and they map naturally to the heartbeat cycle when reliable beat timing is available. In this work, we focus on phase-aligned template learning/cancellation driven by heartbeat timing in three deployment modes: non-causal offline, buffered causal (one-beat latency), and strict real-time causal (zero added latency). The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable.

The rest of this paper is organized as follows. Section II states the signal model and timing-only CFA-removal problem. Section III develops repetitive template cancellation in non-causal and causal forms. Section IV reports experiments on real ear-EEG and semi-synthetic data. Section V summarizes the main findings and practical implications.

II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [8] [9], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. Explicit generative models of the ECG waveform itself [10] are deliberately not required here. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. The R-peak phase-asynchronous disturbances and noise $n(t)$ are assumed zero-mean under the*

Affiliations: ¹Institute of Brain and Mind, CA, USA; ²Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Ontario, Canada; ³University of Oulu, Oulu, Finland; ⁴Synchroni Scientific Instruments Inc., Canada; ⁵Shenzhen Institutes of Advanced Technology, China; ⁶Palos Verdes Peninsula High School, CA, USA; ⁷University of California, Irvine, CA, USA; Corresponding author: Wenlin Zhang (e-mail: william@ibmind.org).

R-peak-synchronous averaging, and DC offset and slow drift are removed by preprocessing.

Assumption 2. The neural signal of interest $s(t)$ is not phase-locked to the *R*-peak: across beats, its phase relative to the cardiac cycle is effectively uniform. Consequently $s(t)$ is preserved by the proposed phase-synchronous repetitive run-out (RRO) cancellation. More detailed analysis is provided in Sec. III.

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the *R*-peak-aligned cardiac phase.

Assumption 3. The beat period T_k is assumed to be accurately locked, and slowly varying due to respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.

Problem 1. Find the repetitive cardiac artifact template $\hat{d}_r(\phi(t))$ such that subtracting it from the measured signal recovers the neural signal of interest:

$$\hat{s}(t) = y(t) - \hat{d}_r(\phi(t)) \approx s(t). \quad (4)$$

Under Assumptions 1–3, $\hat{d}_r(\phi(t))$ can be estimated from the *R*-peak-locked structure of $y(t)$ alone.

Note 1. Since $d_r(\phi(t))$ is periodic in the cardiac phase $\phi(t)$ indexed solely by *R*-peak instants $\{t_k\}$, Problem 1 depends only on *R*-peak timing and never on the ECG waveform shape. Consequently, any wearable heartbeat source—smartwatch ECG, wrist PPG, or any device delivering beat timing via a synchronization mechanism—is sufficient to instantiate the solution.

Note 2. In reality, $s(t)$ may contain a phase-synchronous part that has non-zero mean value across beats, e.g. the heartbeat-evoked potential (HEP)—a genuine cortical response time-locked to the cardiac cycle [11]. This HEP component may or may not be absorbed into $\hat{d}_r$. In this paper, we make no claim that the solution of Eq. 4 preserves the phase-synchronous part of the neural signals, and this is a meaningful extension worthy of further investigation.

To instantiate Problem 1 on the discrete sample grid, let $\hat{t}_k$ be the estimated k -th *R*-peak sample instant and f_s the sampling rate. The k -th beat's *R*-peak-aligned $d_r(\phi(t))$ at integer offset p is defined as

$$d_r^{(k)}\left(\hat{t}_k + \frac{p}{f_s}\right) \triangleq d_r^{(k)}(\phi_p), \quad (5)$$

with the discrete phase index and offset range

$$\phi_p \triangleq \phi\left(\hat{t}_k + \frac{p}{f_s}\right) = \frac{p}{f_s T_k}, \quad (6a)$$

$$p \in \{-n_{\text{pre}}, \dots, -1, 0, 1, \dots, n_{\text{post}}\}. \quad (6b)$$

Note 3. In Eq. (6a), we will drop the explicit T_k dependence in ϕ_p as T_k is slowly varying by Assumption 3, and the phase grid p in Eq. (6b) is approximately shared across beats.

Note 4. The template window spans $n_t = n_{\text{pre}} + n_{\text{post}} + 1$ samples ($p = 0$ at the *R*-peak). A practical choice is $n_{\text{pre}} = \lfloor 0.05 f_s \rfloor$ and $n_{\text{post}} = \lfloor 0.55 f_s \rfloor$, covering QRS onset and the full QRS-T complex by 600ms total duration.

III. REPETITIVE DISTURBANCE REJECTION

A. Non-Causal Cancellation

For offline CFA removal, we could run non-causal averaging to obtain the solution of Problem 1. That is, for a fixed phase index p (offset from the *R*-peak), the template value is the average, across N_r beats, of the EEG sampled at the *same* offset p from each detected *R*-peak $\hat{t}_k$:

$$\hat{d}_r(p) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(p), \quad y_k(p) \equiv y(\hat{t}_k + p/f_s). \quad (7)$$

Substituting the signal $y_k(p)$ by its definition from the discrete time version of Eq. (1), we obtain

$$\hat{d}_r(p) = \underbrace{d_r(p)}_{\text{same every beat}} + \frac{1}{N_r} \sum_{k=1}^{N_r} s_k(p) + \frac{1}{N_r} \sum_{k=1}^{N_r} n_k(p). \quad (8)$$

In Eq. (7), we use $\hat{d}_r(p)$ to denote $\hat{d}_r(\phi_p)$ for better readability, and so are the p -indexed notations used in the subsequent discussion.

Taking the expectation of Eq. (8) over *R*-peaks, we obtain

$$\mathbb{E}[\hat{d}_r(p)] = d_r(p) + \mathbb{E}[s_k(p)] + \mathbb{E}[n_k(p)]. \quad (9)$$

The repetitive *R*-peak-locked disturbance template $d_r(p)$ is deterministic. Ideally, the non-*R*-peak-locked neural term $\mathbb{E}[s_k(p)]$ and the noise term $\mathbb{E}[n_k(p)]$ vanish by Assumption 1, so that Eq. (8) yields the desired estimate by design:

$$\hat{d}_r(p) \approx d_r(p), \quad (10)$$

The CFA removed EEG signal is approximated by

$$\hat{s}_k(p) \approx y_k(p) - \hat{d}_r(p), \quad (11)$$

Let the asynchronous part at offset p have per-sample variance

$$\text{Var}[s_k(p) + n_k(p)] = V(p). \quad (12)$$

The noise floor leakage in $\hat{d}_r(p)$ due to limited sample average is the sample-mean of these N_r terms. If the per-beat samples are *mutually uncorrelated*,

$$\text{Var}[\hat{d}_r(p)] = \frac{1}{N_r^2} \sum_{k=1}^{N_r} V(p) = \frac{V(p)}{N_r}. \quad (13)$$

Realistically, part of the neural background is weakly coupled to the cardiac cycle; retaining the cross terms inflates Eq. (13) by a factor $1 + \delta_{\text{coh}}$, with $\delta_{\text{coh}} = (2/N_r) \sum_{i < j} \rho_{ij}(p)$ and $\rho_{ij}(p)$ the correlation between beats i and j at offset p . Provided δ_{coh} stays bounded, the leakage still decays as $1/N_r$, and Sec. IV-A measures the resulting floor directly with a phase-scramble null.

B. Causal Cancellation

For streaming operation the batch average of Eq. (7) is replaced by its standard exponentially-weighted ILC recursion, updated once per learnable beat at each offset p :

$$\hat{d}_r^{(k)}(p) = \hat{d}_r^{(k-1)}(p) + a [y_k(p) - \hat{d}_r^{(k-1)}(p)], \quad (14a)$$

$$\hat{s}_{k-1}(p) \approx y_{k-1}(p) - \hat{d}_r^{(k-1)}(p). \quad (14b)$$

where $a \in (0, 1]$ is a tunable learning gain: Eq. (14a) is a first-order beat-domain filter with pole $1 - a$, trading an effective averaging span of about $1/a$ beats against a steady-state leakage of $a/(2-a)$ times $V(p)$. The supervisor freezes the update on premature/missed beats. Eq. (14b) is the buffered one-R-peak-delay cancellation form, equivalent to Eq. (11) with index $k - 1$. For true strict real-time cancellation, the current beat's R-peak t_k has not arrived yet, so the canceller must anchor phase using the PI-type R-peak phase tracker prediction rather than detected t_k timing; this enables zero added latency at the cost of prediction-jitter-limited suppression around the R spike. Both modes are $O(1)$ per sample with $O(n_t)$ memory, which makes the proposed method favorable for real-time implementation.

Note 5. *Real-time cancellation requires that the R-peak detector has basic capability to reject immature beats, or to invalidate abnormal R-peaks, so that both template update and cancellation can be held when the timing reference is unreliable. We provide a functional robust R-peak detector and phase tracker for data analysis in the next subsection; however, detailed detector/tracker design is outside the scope of this paper.*

IV. EXPERIMENTS

To evaluate the method we implemented a robust R-peak detector and PI phase tracker for both real and simulated data [12]. We demonstrate the RRO-template canceller only on records where our detector achieves reliable phase locking, and defer general-purpose phase tracking to future work. We likewise omit the strict-causal canceller below, as it hinges on timely R-peak prediction rather than on the template itself.

A. Real Ear-EEG Validation

We use the two-channel ear-EEG configuration of the SeizeIT2 dataset, a multicenter wearable epilepsy corpus of long-duration multimodal recordings (behind-the-ear EEG, ECG, EMG, ACC, GYR) with focal-seizure annotations across a large cohort [13]. Each EEG channel is high-pass filtered at 0.5 Hz, notch filtered at 50 Hz and resampled onto the ECG sample grid, and only supervisor-approved learnable beats enter the average. Because the 600 ms window can exceed the beat interval at high heart rates, overlapping template contributions are averaged before subtraction.

Fig. 1 shows the result of non-causal repetitive disturbance rejection to the SeizeIT2 segment: over a 600 s window the template is averaged across $N_r = 851$ learnable beats, and the qualitative view of this figure shows the repetitive structure of the CFA removal by the proposed method.

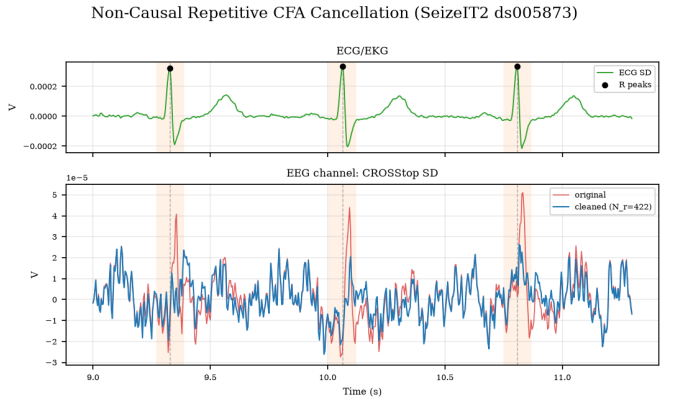


Fig. 1. Non-causal repetitive CFA cancellation on a short SeizeIT2 segment (ds005873, sub-070, ses-01, run-20, behind-the-ear channel). Top: chest ECG (green) with detected R-peaks (black markers). Bottom: original (red) and non-causally cleaned (blue) ear-EEG. The repetitive template $\hat{d}_r(p)$ of Eq. (7) is averaged over the learnable beats of the analysed segment and subtracted as in Eq. (11).

Template SNR via a phase-scramble null. Since the true disturbance d_r is unobservable on real recordings, we validate the template against a null that shares its statistics but destroys R-phase locking, rather than against a direct error to d_r . Define the R-locked energy of a signal z under a peak set P ,

$$\mathcal{E}_P[z] = \left\| \frac{1}{|P|} \sum_{k \in P} z(t_k + p/f_s) \right\|_2^2, \quad (15)$$

evaluated over the main-lobe window around the R-peak (50 ms pre, 150 ms post, L_{lobe} samples), where the R-locked deflection is concentrated. Averaging on the true peaks gives $\hat{d}_r = d_r + \ell$, with leakage ℓ of scale $\text{SD}/\sqrt{N_r}$ by Eq. (13); a circular time-shift of the peak train, which preserves the RR-interval sequence but destroys absolute phase, gives leakage only. Averaging 200 such shifts yields the null floor $\overline{\mathcal{E}_{\text{scr}}}$. Their ratio,

$$Q = \frac{\mathcal{E}_{\text{true}}[y]}{\overline{\mathcal{E}_{\text{scr}}}[y]}, \quad \text{SNR}_{\text{tmpl}} = 10 \log_{10} Q, \quad (16)$$

is the CFA energy standing above the RRO-removed noise floor; since the same leakage enters both numerator and denominator, Q estimates $1 + \text{SNR}$, an offset negligible at the levels below. We further define a bias-corrected CFA RMS, a scale-free measure of the artifact the template removes:

$$A/\text{SD} = \frac{\sqrt{\max(\mathcal{E}_{\text{true}} - \overline{\mathcal{E}_{\text{scr}}}, 0)/L_{\text{lobe}}}}{\text{SD}}. \quad (17)$$

The subtraction removes the leakage bias and $\text{SD} = \text{std}(y)$ is computed over the full preprocessed segment, so A/SD is the R-locked deflection in units of EEG standard deviation. For the real-data example in Fig. 1, the metric reads $\text{SNR}_{\text{tmpl}} = 28.80$ dB and $A/\text{SD} = 1.061$, indicating a strong and clearly resolvable R-locked component. From the whole dataset, we could observe SNR_{tmpl} above 20 dB in 319 runs from 36 subjects, with $A/\text{SD} \in [0.110, 1.220]$ and mean $A/\text{SD} =$

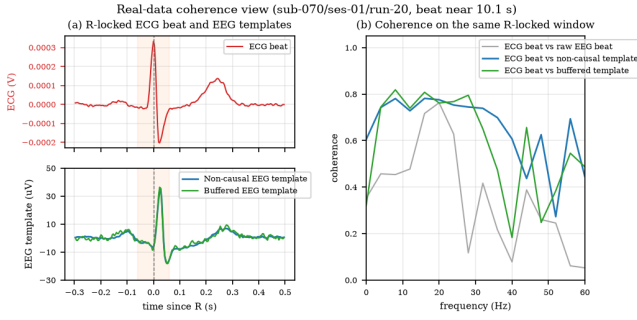


Fig. 2. Coherence view on the same real recording. Left: one R-locked ECG beat overlaid with the non-causal EEG template and the buffered one-beat-delay EEG template. Right: coherence between the windowed ECG beat and three R-locked EEG views: the raw EEG beat, the stacked non-causal template, and the buffered template. Both templates lock to the same R-centered structure and retain appreciable coherence with the ECG at lower frequencies, while the raw beat is diluted by the neural background. Estimated from a single R-locked window, these curves indicate band structure and are not used quantitatively.

0.557, as a showcase of the proposed method’s effectiveness. Again, robust R-peak/beat-phase estimation on arrhythmic or low-quality ECG, and systematic characterization across the full dataset, are outside the scope of this work.

Coherence view. Fig. 2 contrasts the same R-locked ECG window against three EEG views: the raw ear-EEG beat, the stacked non-causal template $\hat{d}_r$, and the buffered one-beat-delay template. The left panel shows that the artifact is timing-repeatable: both templates lock to the same R-centered structure, and the buffered template is nearly indistinguishable from the non-causal one. The right panel shows that the templates also retain appreciable coherence with the ECG beat over the lower part of the band, while the raw-beat coherence is lower and less consistent, diluted by the neural background. Estimated from a single R-locked window, these curves are read as band structure rather than as coherence values.

The consequence is therefore not that the ECG waveform is uninformative, but that exploiting it is more demanding than exploiting timing. An ECG-referenced canceller must be driven by the waveform itself, sample-aligned to the EEG, whereas Eq. (7) consumes only the instants $\{t_k\}$ (Note 1). Both are available on a single instrument; across the device boundary targeted here, only the timing is. Sec. IV-B compares the two under controlled predictability, where ground truth is available.

B. Synthetic Simulation

We now sweep the ECG-to-artifact channel from linear (high coherence) to strongly nonlinear (near-zero coherence) at fixed artifact power. ICA-based CFA removal is effective given sufficiently many channels but not in low-channel-count earables, so we compare against a standard RLS reference-based canceller. Fig. 3 compares the linear ECG-referenced baseline (RLS-ANC) against the proposed rejection under controlled predictability. RLS-ANC acts as a coherence-gated canceller, suppressing the CFA only to the noise floor and degrading as ECG-to-CFA coherence falls. The repetitive methods remain

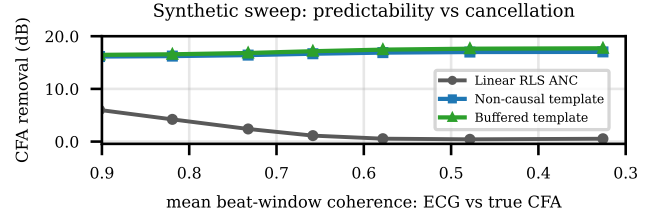


Fig. 3. Synthetic sweep of reference predictability versus cancellation performance. The sweep blends an ECG-coherent linear artifact with a phase-locked but waveform-decorrelated component at fixed artifact power; the x-axis is the mean 1–20 Hz coherence between the averaged ECG beat and the averaged *true* CFA beat, and the y-axis is SNR improvement over the contaminated input measured against the true neural signal. Linear RLS-ANC tracks coherence, whereas the non-causal and buffered template methods stay effective by relying on timing repeatability rather than waveform predictability.

stable, since they estimate the heartbeat-locked disturbance directly from the EEG using beat timing, and so remove nonlinear and coherence-diluted R-locked components as well as the linearly coherent part. Note that RLS-ANC is granted a sample-aligned ECG waveform throughout, i.e. its best case; the repetitive canceller matches it without requiring one.

V. CONCLUSION

We have presented a control-theoretic view of cardiac-field-artifact (CFA) removal in low-channel ear-EEG: CFA is modeled as a heartbeat-phase-locked repetitive disturbance and rejected by phase-indexed synchronous averaging and iterative learning, anchored only to R-peak timing and never to the ECG waveform, so the reference may originate from a separate device that shares only beat instants. By improving SNR in the low-channel ear-EEG setting, it supports non-invasive wearable BCI applications such as brain-state monitoring and low-channel whole-brain modeling [14].

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. This perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact by the available sensors.

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